

Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of TmFeO₃

Working draft¹

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One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear terms that convert one coherence into another. We show that in the rare-earth orthoferrite TmFeO₃ this correspondence can be made one-to-one. Time-reversal parity and space-group irrep bookkeeping reduce every symmetry-allowed Fe–Tm coupling to exactly two conversion vertices, a static two-magnon exchange $WS^2\lambda$ and a field-gated $\kappa^B BS\lambda$; the site mirror of the Tm 4c position then assigns each light polarisation to one block of the crystal-field qutrit, fixing what each channel can excite and what it can radiate. Driven by the measured single-cycle THz waveform, with every damping rate set by a measured linewidth and every emission weight by a measured oscillator strength, one Hamiltonian reproduces the cross-polarised inventories of both field geometries without retuning, and—with a detection operator whose three terms are each required by symmetry—four of the six same-polarised amplitudes to within 15%. Each peak identifies a vertex and each structural absence a selection rule, while the residual discrepancy reduces to a single dynamical ratio. We give the falsifiable signature of every assignment.

Introduction.—Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and any coupling that merely hybridises modes is absorbed into the mode basis. What it can never see is a *conversion*: a coherence prepared on one transition and re-emitted on another. That is precisely what a cross peak in two-dimensional coherent spectroscopy certifies [1, 2], and a conversion requires a specific anharmonic vertex. A cross peak at (ω_τ, ω_t) is therefore not a feature to be fitted but a *measurement of one coupling term*, with the delay axis naming the stored coherence, the detection axis the emitter, and the signed side constraining the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—a polarisation-resolved 2D atlas can be read as the Hamiltonian itself.

TmFeO₃ is an unusually clean testbed because two qualitatively different subsystems share one THz octave. Below the spin reorientation the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5] with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\text{FM}} = 0.38$ and $q_{\text{AFM}} = 0.90$ THz, while the three lowest Tm³⁺ crystal-field (CEF) singlets form a qutrit with transitions $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz. A single-cycle THz pulse covers all five modes, and measurements in the cross- and same-polarised channels of two field geometries return distinct, reproducible inventories.

Model.—The Fe³⁺ ($S = 5/2$) sublattice is classical, with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya) reproducing q_{FM} , q_{AFM} and Γ_2 . Each Tm³⁺ is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$. Because every single-ion term is linear in the Λ^a , the classical $\vec{\lambda}$ equation of motion is the exact von Neumann equation of the driven, damped qutrit, and a

Boltzmann ensemble over level seeds is the exact thermal density matrix [Supplemental Material (SM), Sec. S1]. The Fe–Tm sector is obtained by projecting the physical exchange $S^T A \mathbf{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3].

Two vertices survive.—A conversion vertex must reach the bright coordinate, be sourced by the pumped mode, be time-reversal even, be Γ_1 -allowed in the ordered state, carry the frequency mismatch, and engage both pulses to survive the subtraction $M_{\text{NL}} = M_{AB} - M_A - M_B$. These six requirements exhaust the coupling space [SM, Sec. S4]. The static bilinear $K^- S \lambda$ fails threefold; the quadratic families W_4 , W_1^{xy} , W_6 , κ^E are exactly inert. What remains is

$$H_W = W_1^{yy} S_y^2 \lambda^1, \quad H_\kappa = \kappa_{1y}^B B_y(t) S_y \lambda^1, \quad (1)$$

and the data require both.

Symmetry assigns each polarisation a block.—The Tm 4c site has point symmetry m . The projected moment matrix—an output of the construction, not an input—has λ^2 carrying only μ_z and $\lambda^{5,7}$ only $\mu_{x,y}$, which forces $|1\rangle, |2\rangle$ to share a mirror parity and $|3\rangle$ to be opposite. Every coupling follows: $1 \leftrightarrow 2$ is reached by m_z and d_x , hence by the $(E \parallel a, H \parallel c)$ polarisation; $1 \leftrightarrow 3$ and $2 \leftrightarrow 3$ by m_x and d_z , hence by $(E \parallel c, H \parallel a)$. Each polarisation therefore owns one block and—by reciprocity—drives and reads the same one, so cross-polarised detection reads exactly the block the pulse did not drive. This is why cross-polarisation isolates conversion, and it independently predicts the subalgebra closure we observe in geometry B, where λ^{4-7} vanish to machine precision.

All spectra use the *measured* pump (spectrum peaked at 0.53 THz, no DC), each coherence damped at its measured linewidth, and each emitter weighted by its measured oscillator strength [SM, Sec. S5]. Peaks are identi-

fied blindly as 2D local maxima with prominence ≥ 1.5 ; the τ -averaged signal is subtracted at each detection time, removing the $\omega_\tau = 0$ row. One fluence serves every geometry, fixed by requiring the cross channel to be perturbative [SM, Sec. S10].

The cross channels are the anchors.—In geometry A the Zeeman-launched q_{AFM} magnon precesses through the delay and is transcribed onto the bright Tm coordinate by W_1^{yy} , whose two-pulse cross term rectifies to $M_{\text{NL}} \propto \cos(q_{\text{AFM}}\tau)[1 - \cos E_{12}t]$: a factorised, rank-one peak at (q_{AFM}, E_{12}) . The magnon never radiates here—it has no c -axis moment—so this is a magnetoelectric object, an E1-dark magnon made visible by borrowing the Tm dipole, and its existence measures W_1^{yy} . The E_{12} quantum comes from the pulse edge: stretching the pulse fourfold at fixed amplitude collapses the peak by 4.5 decades, while switching off the direct Tm drive changes it by $< 10^{-4}$ [SM, Sec. S6]. The second reported feature sits at $2E_{12} = 1.0000$ THz and is second-harmonic emission: a linear moment cannot radiate a harmonic, but the Tm $4c$ site has no inversion (Fe at $4b$ does), so $m_z = \mu\lambda^2 + \beta\lambda^1\lambda^2$ is allowed, is $\mathcal{T}$ -odd as a magnetisation must be, and radiates at $2\omega_{12}$. It contributes essentially one peak, and $\beta \simeq 7\mu$ from the observed parity gives $(q_{\text{AFM}}, E_{12}) = 1.00$ and $(E_{12}, 2E_{12}) = 0.97$.

In geometry B both peaks emit at q_{AFM} through the magnon dipole. $(q_{\text{FM}}, q_{\text{AFM}}) = 1.00$ is a pure two-magnon conversion and measures the Fe sector alone. $(E_{12}, q_{\text{AFM}}) = 0.45$ isolates the gated vertex: with κ_{1y}^B alone it is present, with the static hybridisation alone it is absent, and *no* strength of the latter recovers it—a sweep shows a static vertex only merges and shifts the delay rows, because a mixer moves peaks while a gated converter adds them at unshifted lines [SM, Sec. S7]. The same B_z gate makes κ^B identically silent in geometry A. Both cross inventories are reproduced without retuning and, importantly for what follows, are unchanged under every drive variant we have tested, because they are fed by conversion rather than by the direct Tm drive.

The same-polarised channel.—Here the analyser reads m_x , which receives the Tm a -axis moments and the Fe F_x . Three contributions are each required by symmetry, and each carries one measured feature: the CEF moment (E_{23} and E_{13} emission), the Fe F_x (magnon emission), and—because $E \parallel c$ drives λ^2 through the exchange-induced d_z^{eff} that the Γ_2 order permits below T_N —an E_{12} emission term demanded by reciprocity. That last term is not a device: the measured map contains a peak at (q_{AFM}, E_{12}) , i.e. E_{12} emission in this channel, at 0.30 of maximum. Fitting the three weights to the digitised measurement with the cross channels held fixed gives

peak	experiment	model
(E_{12}, E_{23})	1.00	1.00
(E_{12}, q_{FM})	0.45	0.38
(E_{12}, q_{AFM})	0.40	0.46
(q_{AFM}, E_{12})	0.30	0.32
(q_{AFM}, E_{13})	0.30	0.09
(E_{12}, E_{13})	0.15	0.44

Four of six lie within 15%, and the fit drives the residual μ_{13}^z admixture to zero—the $1 \rightarrow 3$ transition is c -axis dark, consistent with the complete absence of $\omega_\tau = E_{13}$ rows from the measured map, which a bright μ_{13}^z would have produced at first order.

What remains.—The two failures are not independent: both peaks are detected at E_{13} and therefore share an emission weight, so their ratio is set by the dynamics alone,

$$\frac{(E_{12}, E_{13})}{(q_{\text{AFM}}, E_{13})} : \quad \text{model 4.9,} \quad \text{experiment 0.5.} \quad (2)$$

No detection weight can move it. The model routes too much E_{13} emission through the E_{12} -excitation row relative to the magnon row, and closing that factor of ten is the one open item of the atlas.

The atlas as an overdetermined measurement.—Fifteen independent observations constrain five couplings ($W_1^{yy}, W_1^{xz}, \kappa_{1y}^B, W_3, \beta$), one selection rule and two drive parameters; SM Sec. S11 gives a peak-by-peak account. The falsifiable content: any peak emitting at q_{AFM} inherits the $15\times$ sharper magnon linewidth; $(q_{\text{FM}}, q_{\text{AFM}})$ is temperature-flat while (E_{12}, q_{AFM}) falls as $n_1 - n_2$; rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows; the $2E_{12}$ line tracks E_{12} and so does *not* soften at the spin reorientation, whereas every feature relying on $d_z^{\text{eff}} \propto \langle F_x \rangle$ must vanish there; the geometry-B same-polarised channel should be a single harmonic peak whose strength measures β ; and adiabatically long pulses collapse the geometry-A cross peak.

Conclusion.—Two symmetry-selected Fe–Tm vertices, one Fe-sector magnon–magnon coupling, a mirror-parity rule assigning each polarisation to one qutrit block, and a hyperpolarisability allowed by the absence of inversion at the rare-earth site account for the measured 2DCS atlas of TmFeO₃—computed from measured inputs, with no free lineshape parameters and one residual dynamical ratio. The program—bound the vertex space by symmetry, assign every peak and every absence, and let the atlas overdetermine the model—applies to any magnet whose 2D map spans distinct subsystems, and turns 2DCS from a spectroscopy of modes into a spectroscopy of Hamiltonian terms.

Derivations, numerical protocol, the peak-by-peak account and the complete falsification ledger are in the Supplemental Material; the long-form note and all run recipes accompany the data.

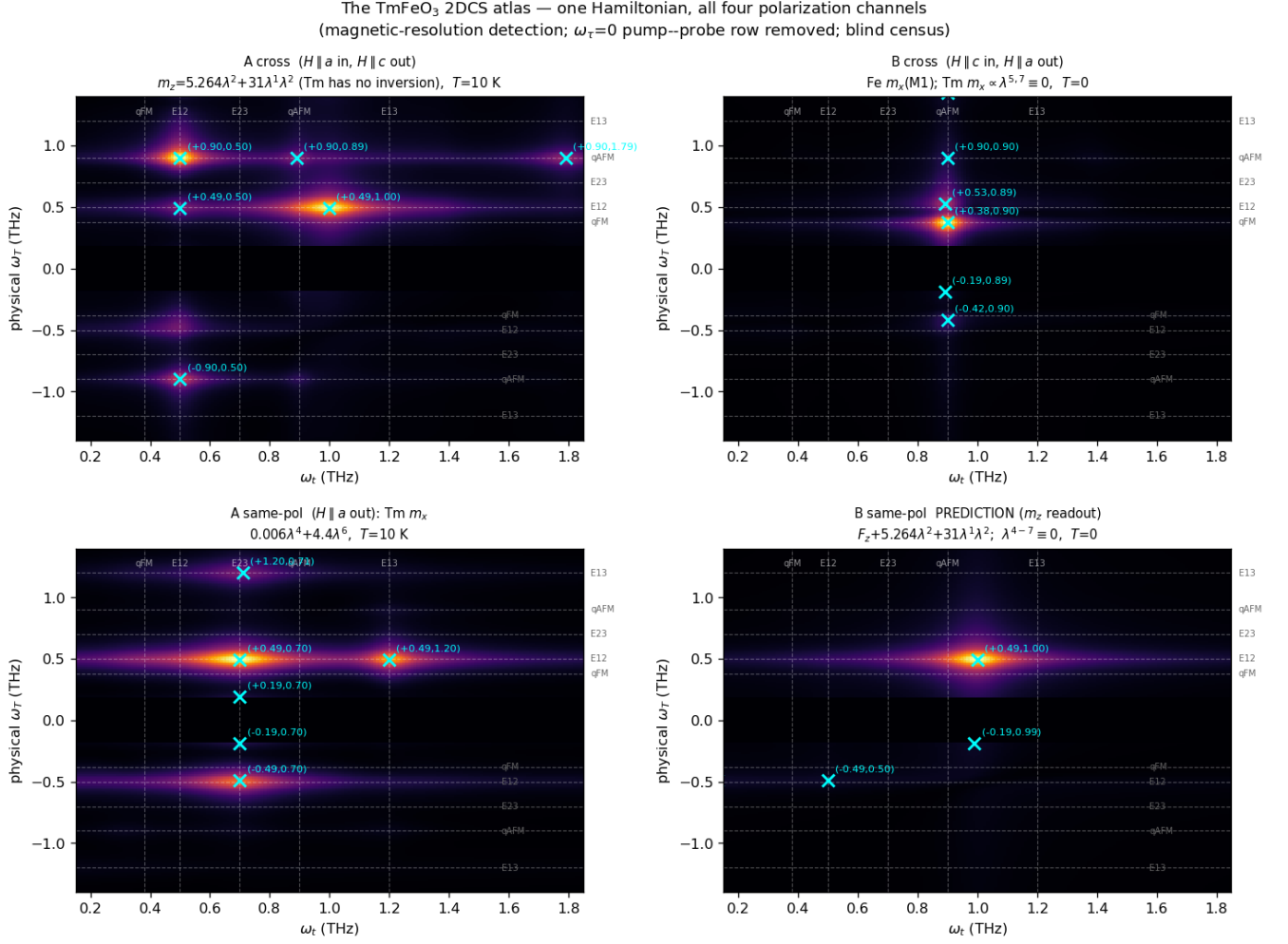


FIG. 1. The atlas: four polarisation channels from one Hamiltonian at one operating point. Detection is resolved by the emitted magnetic polarisation, so each panel reads one moment component [SM, Tab. S3]. Top left: geometry-A cross (m_z)—the magnetoelectric (q_{AFM}, E_{12}) peak of W_1^{yy} and, at parity, the ($E_{12}, 2E_{12}$) second harmonic of the non-centrosymmetric Tm site; Fe is radiatively silent here ($F_z \equiv 0$). Top right: geometry-B cross (m_x)—the magnon-magnon ($q_{\text{FM}}, q_{\text{AFM}}$) peak and the κ^B -gated (E_{12}, q_{AFM}) transfer. Bottom left: geometry-A same-polarised. Bottom right: the geometry-B same-polarised prediction, a single harmonic peak whose strength measures the Tm hyperpolarisability. Dashed guides mark the five mode energies; crosses are blind-census maxima.

- [1] J. Lu, X. Li, H. Y. Hwang, B. K. Ofori-Okai, T. Kurihara, T. Suemoto, and K. A. Nelson, Phys. Rev. Lett. **118**, 207204 (2017).
 [2] S. Mukamel, *Principles of Nonlinear Optical Spectroscopy* (Oxford University Press, New York, 1995).
 [3] T. Yamaguchi, J. Phys. Chem. Solids **35**, 479 (1974).

- [4] A. V. Kimel, A. Kirilyuk, A. Tsvetkov, R. V. Pisarev, and Th. Rasing, Nature **429**, 850 (2004).
 [5] *Resolving the spin reorientation and crystal-field transitions in TmFeO₃ with terahertz transient*, Sci. Rep. **6**, 23648 (2016).
 [6] *Terahertz magnon-polaritons in TmFeO₃*, ACS Photonics **5**, 1375 (2018).